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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

NOTES

Subject with code: Electronic Principles and Circuits (BEC303)

Module 4

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Syllabus :

Negative Feedback: Four Types of Negative Feedback, VCVS Voltage gain, Other VCVS Equations, ICVS Amplifier, VCIS Amplifier, ICIS Amplifier (No Mathematical Derivation).

Active Filters: Ideal Responses, First Order Stages, VCVS Unity Gain Second Order Low pass Filters, VCVS Equal Component Low Pass Filters, VCVS High Pass Filters, MFB Bandpass Filters, Bandstop Filters.

Negative Feedback

The negative feedback amplifier is an electronic amplifier in which the feedback energy (Voltage or Current) is out of phase with input signal and thus opposes it. Negative feedback reduces gain. It also reduces distortion, noise and instability. This feedback increases bandwidth and improves input and output impedance.

Four types of negative feedback

The input to a negative-feedback amplifier can be either a voltage or a current. Also, the output signal can be either a voltage or a current. This implies that four types of negative feedback exist.

Table 19-1 Ideal Negative Feedback									
Input	Output	Circuit	z_{in}	z_{out}	Converts	Ratio	Symbol	Type of amplifier	
V	V	VCVS	∞	0	—	V_{out}/V_{in}	A_v	Voltage amplifier	
I	V	ICVS	0	0	i to v	V_{out}/i_{in}	r_m	Transresistance amplifier	
V	I	VCIS	∞	∞	v to i	i_{out}/V_{in}	g_m	Transconductance amplifier	
I	I	ICIS	0	∞	—	i_{out}/i_{in}	A_i	Current amplifier	

Figure 19-1 (a) Voltage-controlled voltage source; (b) current-controlled voltage source.

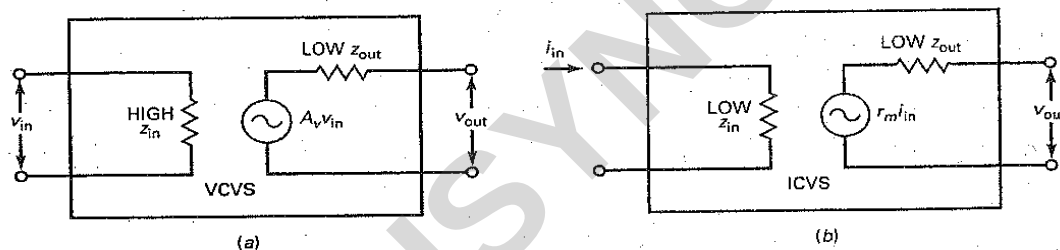
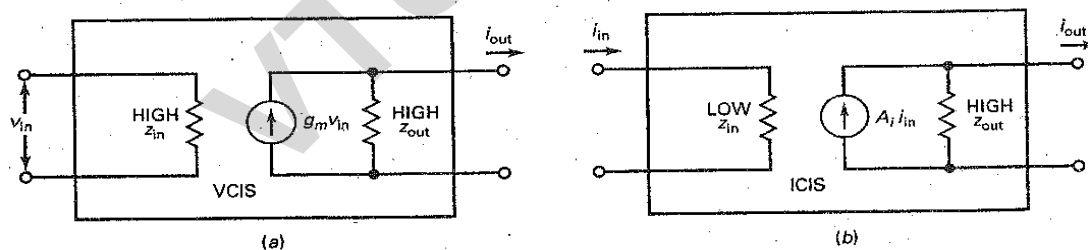


Figure 19-2 (a) Voltage-controlled current source; (b) current-controlled current source.



Voltage controlled voltage source (VCVS)

As shown in Table 19-1, the first type has an input voltage and an output voltage. The circuit that uses this type of negative feedback is called a **voltage-controlled voltage source (VCVS)**. A VCVS is an ideal voltage amplifier because it has a stabilized voltage gain, infinite input impedance, and zero output impedance as shown.

Figure 19-1a shows the VCVS, a voltage amplifier. With practical circuits, the input impedance is not infinite, but it is very high. Likewise, the output impedance is not zero, but it is very low. The voltage gain of the VCVS is symbolized A_v . Since z_{out} approaches zero, the output side of a VCVS is a stiff voltage source to any practical load resistance.

Current controlled voltage source (ICVS)

In the second type of negative feedback, an input current controls an output voltage. The circuit using this type of feedback is called a **current-controlled voltage source (ICVS)**. Because an input current controls an output voltage, an ICVS is sometimes called a **transresistance amplifier**. The word *resistance* is used because the ratio of v_{out}/i_{in} has the unit of ohms. The prefix *trans* refers to taking the ratio of an output quantity to an input quantity.

Figure 19-1b shows an ICVS, a transresistance amplifier (current-to-voltage converter). It has a very low input impedance and a very low output impedance. The conversion factor of the ICVS is called *transresistance*, symbolized r_m and expressed in ohms.

Voltage controlled current source (VCIS)

The third type of negative feedback has an input voltage controlling an output current. The circuit using this type of negative feedback is called a **voltage-controlled current source (VCIS)**. Because an input voltage controls an output current, a VCIS is sometimes called a **transconductance amplifier**. The word *conductance* is used because the ratio of i_{out}/v_{in} has the unit of siemens (mhos).

Figure 19-2a shows a VCIS, a transconductance amplifier (voltage-to-current converter). It has a very high input impedance and a very high output impedance. The conversion factor of the VCIS is called *transconductance*, symbolized g_m and expressed in siemens (mhos).

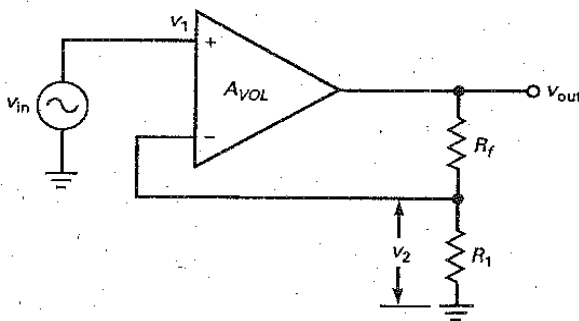
Current controlled current source (ICIS)

In the fourth type of negative feedback, an input current is amplified to get a larger output current. The circuit with this type of negative feedback is called a **current-controlled current source (ICIS)**. An ICIS is an ideal current amplifier because it has a stabilized current gain, zero input impedance, and infinite output impedance.

Figure 19-2b shows an ICIS, a current amplifier. It has very low input impedance and very high output impedance. The current gain of the ICIS is symbolized A_i . Since z_{out} approaches infinity, the output side of a VCVS is a stiff current source to any practical load resistance.

VCVS voltage gain

Figure 19-3 VCVS amplifier.



Exact closed loop voltage gain

Figure 19-3 shows a noninverting amplifier. The op amp has an open-loop voltage gain of A_{VOL} , typically 100,000 or more. Because of the voltage divider, part of the output voltage is fed back to the inverting input. The **feedback fraction B** of any VCVS circuit is defined as feedback voltage divided by the output voltage. In Fig. 19-3:

$$B = \frac{v_2}{v_{out}} \quad (19-1)$$

The feedback fraction is also called the **feedback attenuation factor** because it indicates how much the output voltage is attenuated before the feedback signal reaches the inverting input.

With some algebra, we can derive the following exact equation for the closed-loop voltage gain:

$$A_{v(CL)} = \frac{A_{VOL}}{1 + A_{VOL} B} \quad (19-2)$$

or with the notation of Table 19-1, where $A_v = A_{v(CL)}$:

$$A_v = \frac{A_{VOL}}{1 + A_{VOL} B} \quad (19-3)$$

This is the exact equation for the closed-loop voltage gain of any VCVS amplifier.

Ideal closed loop voltage gain

For a VCVS to work well, the loop gain $A_{VOL}B$ must be much greater than unity. When the designer satisfies this condition, Eq. (19-3) becomes:

$$A_v = \frac{A_{VOL}}{1 + A_{VOL} B} \cong \frac{A_{VOL}}{A_{VOL} B}$$

or

$$A_v \cong \frac{1}{B} \quad (19-4)$$

Using ideal equation voltage gain for noninverting amplifier of fig 19.3

All you have to do is calculate the feedback fraction with Eq. (19-1) and take the reciprocal. For instance, in Fig. 19-3, the feedback fraction is:

$$B = \frac{v_2}{v_{out}} = \frac{R_1}{R_1 + R_f} \quad (19-6)$$

Taking the reciprocal gives:

$$A_v \cong \frac{1}{B} = \frac{R_1 + R_f}{R_1} = \frac{R_f}{R_1} + 1$$

Other VCVS Equations

Gain Stability

The gain stability depends on having a very low percent error between the ideal and the exact closed-loop voltage gains. The smaller the percent error, the better the stability. The *worst-case error* of closed-loop voltage gain occurs when the open-loop voltage gain is minimum. As an equation:

$$\% \text{ Maximum error} = \frac{100\%}{1 + A_{VOL(min)} B} \quad (19-7)$$

Closed loop input impedance

Figure 19-5a shows a noninverting amplifier. Here is the exact equation for the closed-loop input impedance of this VCVS amplifier:

$$z_{in(CL)} = (1 + A_{VOL}B)R_{in} \parallel R_{CM} \quad (19-8)$$

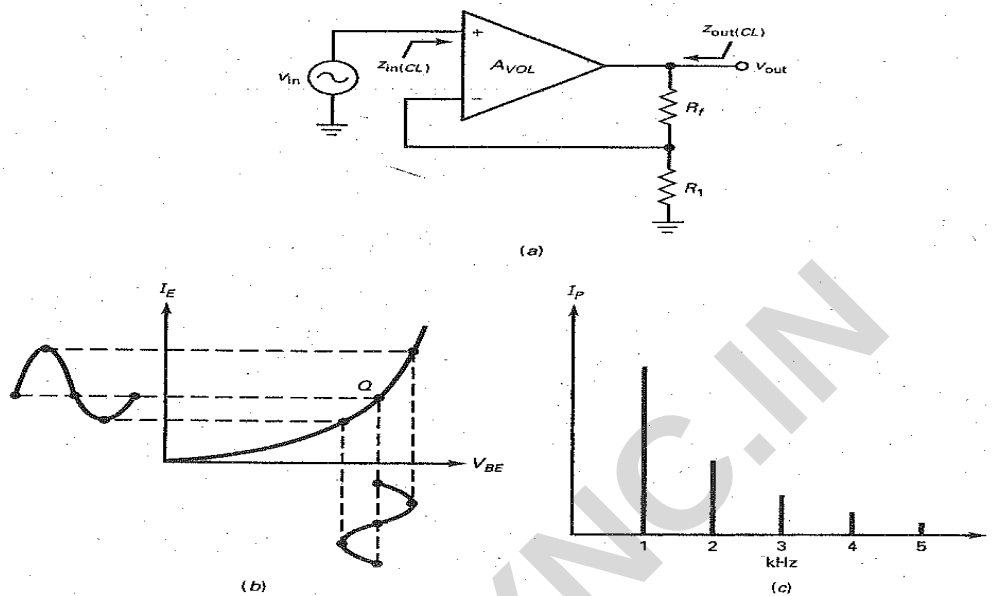
where R_{in} = the open-loop input resistance of the op amp

R_{CM} = the common-mode input resistance of the op amp

Often, R_{CM} is ignored because it is large, and Eq. (19-8) is approximated by:

$$z_{in(CL)} \cong (1 + A_{VOL}B)R_{in} \quad (19-9)$$

Figure 19-5 (a) VCVS amplifier; (b) nonlinear distortion; (c) fundamental and harmonics.



Closed loop output impedance

In Fig. 19-5a, the closed-loop output impedance is the overall output impedance looking back into the VCVS amplifier. The exact equation for this closed-loop output impedance is:

$$z_{out(CL)} = \frac{R_{out}}{1 + A_{VOL}B} \quad (19-10)$$

Nonlinear Harmonic Distortion

One more improvement worth mentioning is the effect of negative feedback on distortion. In the later stages of an amplifier, *nonlinear distortion* will occur with large signals because the input/output response of the amplifying devices becomes nonlinear. For instance, the nonlinear graph of the base-emitter diode distorts a large signal by elongating the positive half cycle and compressing the negative half cycle, as shown in Fig. 19-5b.

Nonlinear distortion produces *harmonics* of the input signal. For instance, if a sinusoidal voltage signal has a frequency of 1 kHz, the distorted output current will contain sinusoidal signals with frequencies of 1, 2, 3 kHz, and so forth, as shown in the *spectrum diagram* of Fig. 19-5c. The fundamental frequency is 1 kHz, and all others are harmonics. The rms value of all the harmonics measured together tells us how much distortion has occurred. This is why nonlinear distortion is often called **harmonic distortion**.

defined as:

percent of total harmonic distortion,

$$THD = \frac{\text{Total harmonic voltage}}{\text{Fundamental voltage}} \times 100\% \quad (19-11)$$

For instance, if the total harmonic voltage is 0.1 V rms and the fundamental voltage is 1 V, then $THD = 10$ percent.

Negative feedback reduces harmonic distortion. The exact equation for closed-loop harmonic distortion is:

$$THD_{CL} = \frac{THD_{OL}}{1 + A_{VOL}B} \quad (19-12)$$

where THD_{OL} = open-loop harmonic distortion

THD_{CL} = closed-loop harmonic distortion

The ICVS Amplifier

Figure 19-7 shows a transresistance amplifier. It has an input current and an output voltage. The ICVS amplifier is an almost perfect *current-to-voltage converter* because it has zero input impedance and zero output impedance.

Voltage Gain

The exact equation for output voltage is:

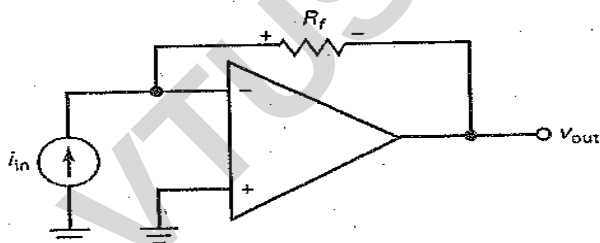
$$v_{out} = -\left(i_{in}R_f \frac{A_{VOL}}{1 + A_{VOL}}\right) \quad (19-13)$$

Because A_{VOL} is much greater than unity, the equation simplifies to:

$$v_{out} = -(i_{in}R_f) \quad (19-14)$$

where R_f is the transresistance.

Figure 19-7 ICVS amplifier.



Input and output impedance

In Figure 19-7, the exact equations for closed-loop input and output impedances are:

$$z_{in(CL)} = \frac{R_f}{1 + A_{VOL}} \quad (19-15)$$

$$z_{out(CL)} = \frac{R_{out}}{1 + A_{VOL}} \quad (19-16)$$

In both equations, the large denominator will reduce the impedance to a very low value.

The VCIS Amplifier

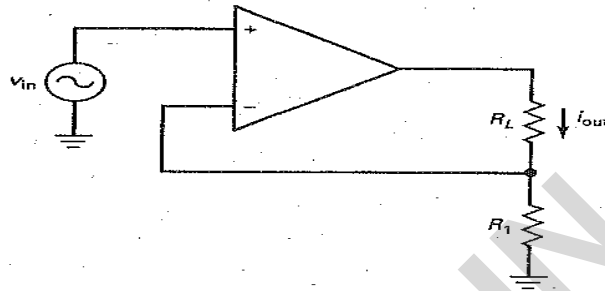
With a VCIS amplifier, an input voltage controls an output current. Because of the heavy negative feedback in this kind of amplifier, the input voltage is converted to a precise value of output current.

Figure 19-10 shows a transconductance amplifier. It is similar to a VCVS amplifier, except that R_L is the load resistor as well as the feedback resistor. In other words, the active output is not the voltage across $R_1 + R_L$; rather, it is the current through R_L . This output current is stabilized; that is, a specific value of input voltage produces a precise value of output current.

In Fig. 19-10, the exact equation for output current is:

$$i_{out} = \frac{v_{in}}{R_1 + (R_1 + R_L)/A_{VOL}} \quad (19-18)$$

Figure 19-10 VCIS amplifier.



In a practical circuit, the second term in the denominator is much smaller than the first and the equation simplifies to:

$$i_{out} = \frac{v_{in}}{R_1} \quad (19-19)$$

This is sometimes written as:

$$i_{out} = g_m v_{in}$$

where $g_m = 1/R_1$.

Input and output impedance of VCIS amplifier

Since the input side of Fig. 19-10 is the same as the input side of a VCVS amplifier, the approximate equation for the closed-loop input impedance of a VCIS amplifier is:

$$z_{in(CL)} = (1 + A_{VOL}B)R_{in} \quad (19-20)$$

where R_{in} is the input resistance of the op amp. The stabilized output current sees a closed-loop output impedance of:

$$z_{out(CL)} = (1 + A_{VOL})R_1 \quad (19-21)$$

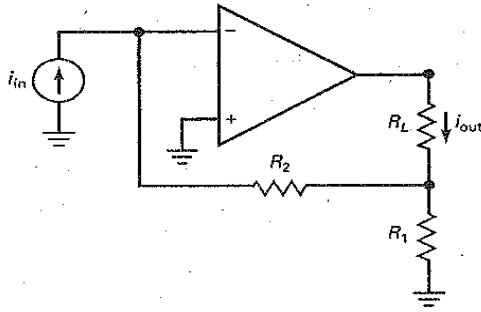
The ICIS Amplifier

An ICIS circuit amplifies the input current. Because of the heavy negative feedback, the ICIS amplifier tends to act like a perfect **current amplifier**. It has a very low input impedance and a very high output impedance.

Figure 19-12 shows an inverting current amplifier. The closed-loop current gain is stabilized and given by:

$$A_i = \frac{A_{VOL}(R_1 + R_2)}{R_L + A_{VOL}R_1} \quad (19-22)$$

Figure 19-12 ICIS amplifier.



Usually, the second term in the denominator is much larger than the first and the equation simplifies to:

$$A_i \cong \frac{R_2}{R_1} + 1 \quad (19-23)$$

The equation for the closed-loop input impedance of an ICIS amplifier is:

$$z_{in(CL)} = \frac{R_2}{1 + A_{VOL}B} \quad (19-24)$$

where the feedback fraction is given by:

$$B = \frac{R_1}{R_1 + R_2} \quad (19-25)$$

The stabilized output current sees a closed-loop output impedance of:

$$z_{out(CL)} = (1 + A_{VOL})R_1 \quad (19-26)$$

A large A_{VOL} produces a very small input impedance and a very large output impedance. Because of this, the ICIS circuit is an almost perfect current amplifier.

Active Filter

A filter is circuit capable of passing certain frequencies while attenuating other frequencies. Thus a filter can extract desired frequency signal and rejects undesirable frequencies. Active filter is type of filter which consists of active element like transistor, vacuum tubes and op-amps along with passive elements like resistors and capacitors.

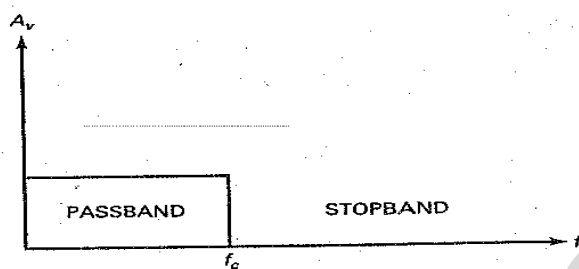
There are 5 types of filters based on their frequency responses (The frequency response is graph of voltage gain vs frequency) as follows.

1. Low pass filter 2. High pass filter 3. Band pass filter
4. Band stop filter 5. All pass filter

Low Pass Filter

Figure 21-1 shows the ideal frequency response of a **low-pass filter**.

Figure 21-1 Ideal low-pass response.



A low-pass filter passes all frequencies from zero to the cutoff frequency and blocks all frequencies above the cutoff frequency.

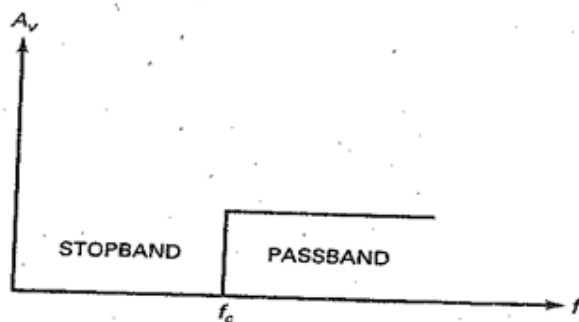
With a low-pass filter, the frequencies between zero and the cutoff frequency are called the **passband**. The frequencies above the cutoff frequency are called the **stopband**. The roll-off region between the passband and the stopband is called the **transition**. An ideal low-pass filter has zero *attenuation* (signal loss) in the passband, infinite attenuation in the stopband, and a vertical transition.

High Pass Filter

Figure 21-2 shows the ideal frequency response of a **high-pass filter**. A high-pass filter blocks all frequencies from zero up to the cutoff frequency and passes all frequencies above the cutoff frequency.

With a high-pass filter, the frequencies between zero and the cutoff frequency are the stopband. The frequencies above the cutoff frequency are the passband. An ideal high-pass filter has infinite attenuation in the stopband, zero attenuation in the passband, and a vertical transition.

Figure 21-2 Ideal high-pass response.



Bandpass Filter

Figure 21-3 shows the ideal frequency response of a bandpass filter. A brick wall response like this blocks all frequencies from zero up to the lower cutoff frequency. Then, it passes all the frequencies between the lower and upper cutoff frequencies. Finally, it blocks all frequencies above the upper cutoff frequency.

With a bandpass filter, the passband is all the frequencies between the lower and upper cutoff frequencies. The frequencies below the lower cutoff frequency and above the upper cutoff frequency are the stopband. An ideal bandpass filter has zero attenuation in the passband, infinite attenuation in the stopband, and two vertical transitions.

The *bandwidth (BW)* of a bandpass filter is the difference between its upper and lower 3-dB cutoff frequencies:

$$BW = f_2 - f_1 \quad (21-1)$$

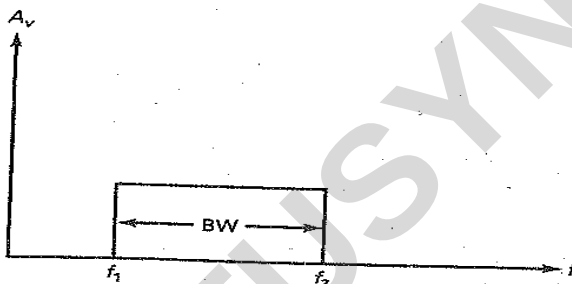
The center frequency is symbolized by f_0 and is given by the geometric average of the two cutoff frequencies:

$$f_0 = \sqrt{f_1 f_2} \quad (21-2)$$

The *Q* of a bandpass filter is defined as the center frequency divided by the bandwidth:

$$Q = \frac{f_0}{BW} \quad (21-3)$$

Figure 21-3 Ideal bandpass response.



If Q is less than 1, the bandpass filter is called a **wideband filter**. If Q is greater than 1, the filter is called a **narrowband filter**. For example, a filter with

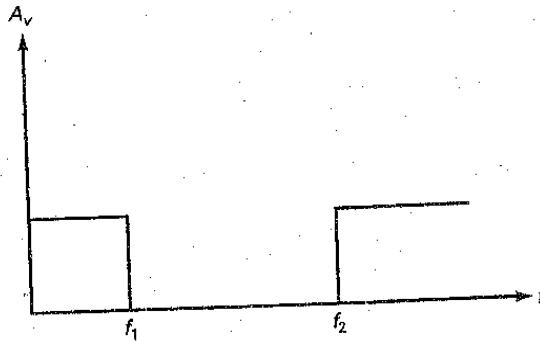
Bandstop Filter

Figure 21-4 shows the ideal frequency response of a **bandstop filter**. This type of filter passes all frequencies from zero up to the lower cutoff frequency. Then, it blocks all the frequencies between the lower and upper cutoff frequencies. Finally, it passes all frequencies above the upper cutoff frequency.

With a bandstop filter, the stopband is all the frequencies between the lower and upper cutoff frequencies. The frequencies below the lower cutoff frequency and above the upper cutoff frequency are the passband. An ideal bandstop filter has infinite attenuation in the stopband, no attenuation in the passband, and two vertical transitions.

The definitions for bandwidth, narrowband, and center frequency are the same as before. In other words, with a bandstop filter, we use Eqs. (21-1) through (21-3) to calculate BW, f_0 , and Q . Incidentally, the bandstop filter is sometimes called a *notch filter* because it notches out or removes all frequencies in the stopband.

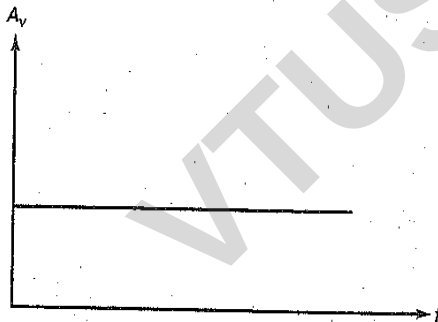
Figure 21-4 Ideal bandstop response.



All Pass filter

Figure 21-5 shows the frequency response of an ideal **all-pass filter**. It has a passband and no stopband. Because of this, it passes all frequencies between zero and infinite frequency. It may seem rather unusual to call it a filter since it has zero attenuation for all frequencies. The reason it is called a filter is because of the effect it has on the *phase* of signals passing through it. The all-pass filter is useful when we want to produce a certain amount of phase shift for the signal being filtered without changing its amplitude.

Figure 21-5 Ideal all-pass response.



First Order Stage(Filter)

First-order or 1-pole active-filter stages have only one capacitor. Because of this, they can produce only a low-pass or a high-pass response. Bandpass and bandstop filters can be implemented only when n is greater than 1.

First Order low pass stage(filter)

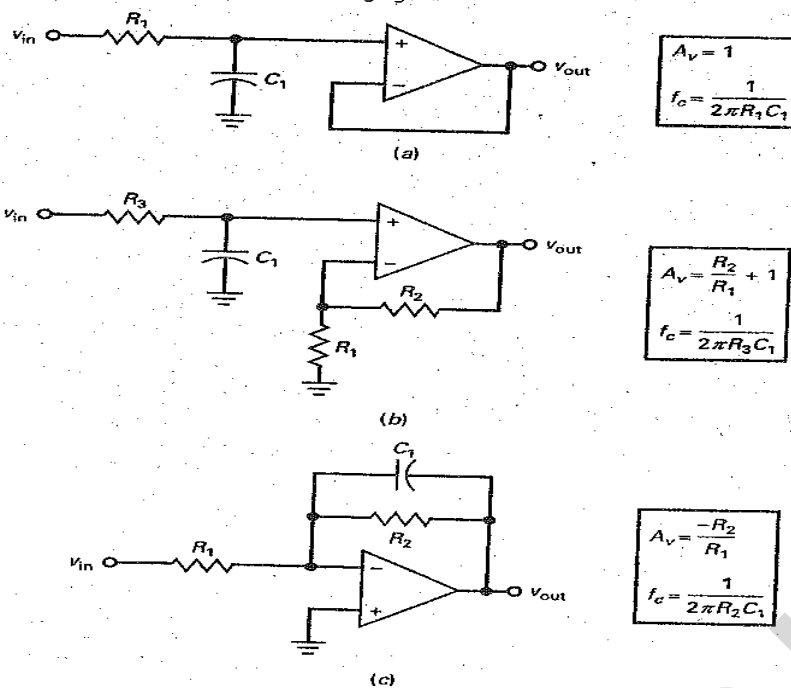
Figure 21-21a shows the simplest way to build a first-order low-pass active filter. It is nothing more than an RC lag circuit and a voltage follower. The voltage gain is:

$$A_v = 1$$

The 3-dB cutoff frequency is given by:

$$f_c = \frac{1}{2\pi R_1 C_1}$$

Figure 21-21 First-order low-pass stages: (a) Noninverting unity gain; (b) noninverting with voltage gain; (c) inverting with voltage gain.



When the frequency increases above the cutoff frequency, the capacitive reactance decreases and reduces the noninverting input voltage. Since the $R_1 C_1$ lag circuit is outside the feedback loop, the output voltage rolls off. As the frequency approaches infinity, the capacitor becomes a short and there is zero input voltage.

Figure 21-21b shows another noninverting first-order low-pass filter. Although it has two additional resistors, it has the advantage of voltage gain. The voltage gain well below the cutoff frequency is given by:

$$A_v = \frac{R_2}{R_1} + 1 \quad (21-10)$$

The cutoff frequency is given by:

$$f_c = \frac{1}{2\pi R_3 C_1} \quad (21-11)$$

Above the cutoff frequency, the lag circuit reduces the noninverting input voltage. Since the $R_3 C_1$ lag circuit is outside the feedback loop, the output voltage rolls off at a rate of 20 dB per decade.

Figure 21-21c shows an inverting first-order low-pass filter and its equations. At low frequencies, the capacitor appears to be open and the circuit acts like an inverting amplifier with a voltage gain of:

$$A_v = -\frac{R_2}{R_1} \quad (21-12)$$

As the frequency increases, the capacitive reactance decreases and reduces the impedance of the feedback branch. This implies less voltage gain. As the frequency approaches infinity, the capacitor becomes a short and there is no voltage gain. As shown in Fig. 21-21c, the cutoff frequency is given by:

$$f_c = \frac{1}{2\pi R_2 C_1} \quad (21-13)$$

First Order High Pass Stages(filter)

Figure 21-22a shows the simplest way to build a first-order high-pass active filter. The voltage gain is:

$$A_v = 1$$

The 3-dB cutoff frequency is given by:

$$f_c = \frac{1}{2\pi R_1 C_1} \quad (21-14)$$

When the frequency decreases below the cutoff frequency, the capacitive reactance increases and reduces the noninverting input voltage. Since the $R_1 C_1$ circuit is

outside the feedback loop, the output voltage rolls off. As the frequency approaches zero, the capacitor becomes an open and there is zero input voltage.

Figure 21-22b shows another noninverting first-order high-pass filter. The voltage gain well above the cutoff frequency is given by:

$$A_v = \frac{R_2}{R_1} + 1 \quad (21-15)$$

The 3-dB cutoff frequency is given by:

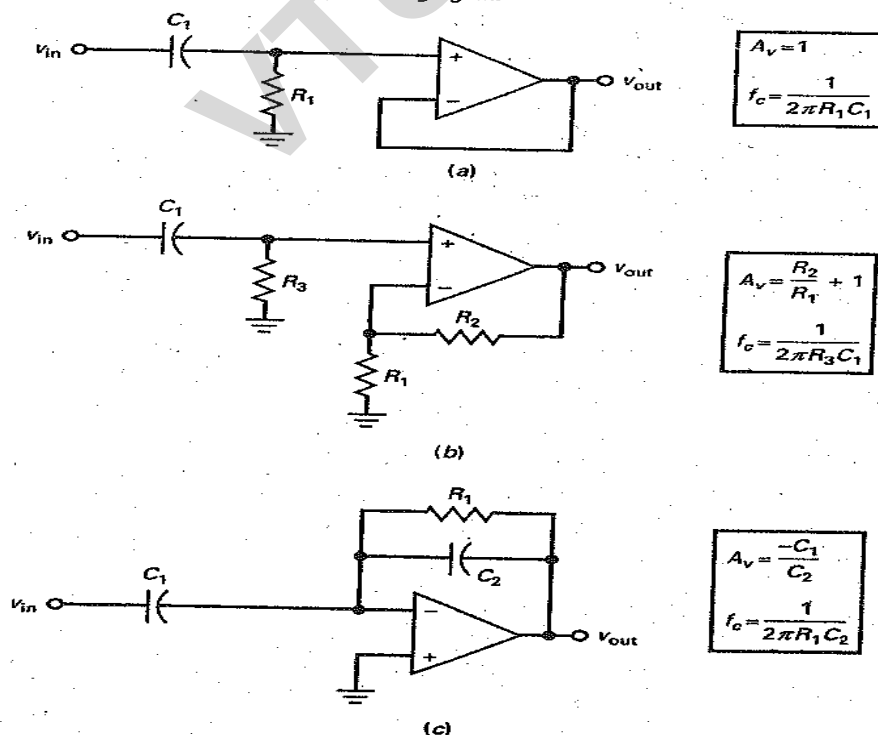
$$f_c = \frac{1}{2\pi R_3 C_1} \quad (21-16)$$

Well below the cutoff frequency, the RC circuit reduces the noninverting input voltage. Since the $R_3 C_1$ lag circuit is outside the feedback loop, the output voltage rolls off at a rate of 20 dB per decade.

Figure 21-22c shows another first-order high-pass filter and its equations. At high frequencies, the circuit acts like an inverting amplifier with a voltage gain of:

$$A_v = \frac{-X_{C2}}{X_{C1}} = \frac{-C_1}{C_2} \quad (21-17)$$

Figure 21-22 First-order high-pass stages: (a) Noninverting unity gain; (b) noninverting with voltage gain; (c) inverting with voltage gain.



As the frequency decreases, the capacitive reactances increase and eventually reduce the input signal and the feedback. This implies less voltage gain. As the frequency approaches zero, the capacitors become open and there is no input signal. As shown in Fig. 21-22c, the 3-dB cutoff frequency is given by:

$$f_c = \frac{1}{2\pi R_1 C_2} \quad (21-18)$$

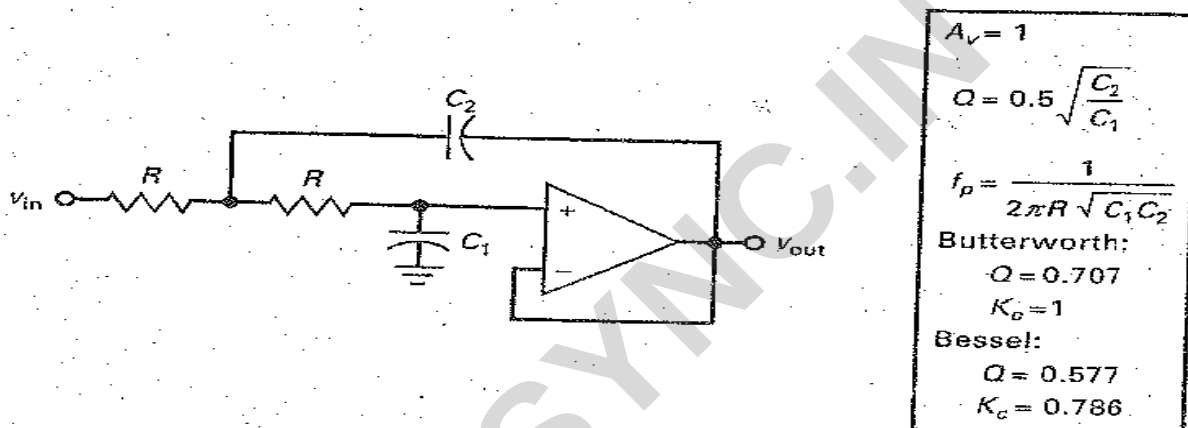
VCVS Unity gain second order low pass filters

Sallen-key second order low pass filter

These filters also called VCVS filters because the op amp is used as a voltage controlled voltage source.

Figure 21-24 shows a Sallen-Key second-order low-pass filter. Notice that the two resistors have the same value, but the two capacitors are different. There is a lag

Figure 21-24 Second-order VCVS stage for Butterworth and Bessel.



circuit on the noninverting input, but this time there is a feedback path through a second capacitor C_2 . At low frequencies, both capacitors appear to be open and the circuit has a unity gain because the op amp is connected as a voltage follower.

As the frequency increases, the impedance of C_1 decreases and the noninverting input voltage decreases. At the same time, capacitor C_2 is feeding back a signal that is in phase with the input signal. Since the feedback signal adds to the source signal, the feedback is *positive*. As a result, the decrease in the noninverting input voltage caused by C_1 is not as large as it would be without the positive feedback.

Butterworth and Bessel response

When analyzing a circuit like the one shown in Fig. 21-24, we start by calculating Q and f_p . If $Q = 0.707$, we have a Butterworth response and a K_c value of 1. If $Q = 0.577$, we have a Bessel response and a K_c value of 0.786. Next, we can calculate the cutoff frequency with:

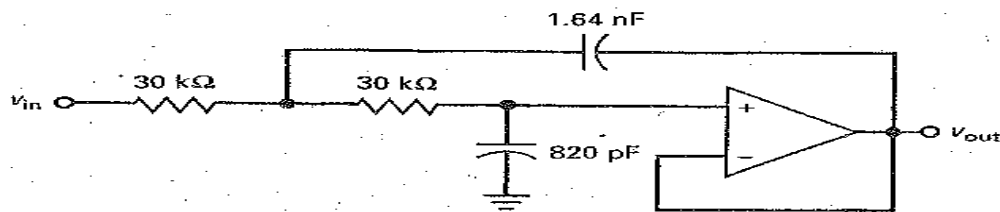
$$f_c = K_c f_p \quad (21-21)$$

Example 21-3

III Multisim

What are the pole frequency and Q of the filter shown in Fig. 21-27? What is the cutoff frequency?

Figure 21-27 Butterworth unity-gain example.



SOLUTION The Q and pole frequency are:

$$Q = 0.5 \sqrt{\frac{C_2}{C_1}} = 0.5 \sqrt{\frac{1.64 \text{ nF}}{820 \text{ pF}}} = 0.707$$

$$f_p = \frac{1}{2\pi R \sqrt{C_1 C_2}} = \frac{1}{2\pi (30 \text{ k}\Omega) \sqrt{(820 \text{ pF})(1.64 \text{ nF})}} = 4.58 \text{ kHz}$$

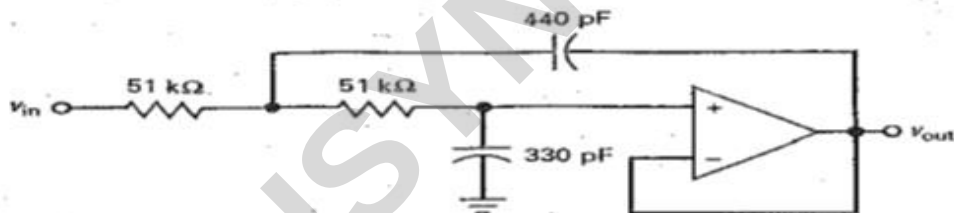
The Q value of 0.707 tells us that this is a Butterworth response, so the cutoff frequency is the same as the pole frequency:

$$f_c = f_p = 4.58 \text{ kHz}$$

Example 21-4

In Fig. 21-28, what are the pole frequency and Q ? What is the cutoff frequency?

Figure 21-28 Bessel unity-gain example.



SOLUTION The Q and pole frequency are:

$$Q = 0.5 \sqrt{\frac{C_2}{C_1}} = 0.5 \sqrt{\frac{440 \text{ pF}}{330 \text{ pF}}} = 0.577$$

$$f_p = \frac{1}{2\pi R \sqrt{C_1 C_2}} = \frac{1}{2\pi (51 \text{ k}\Omega) \sqrt{(330 \text{ pF})(440 \text{ pF})}} = 8.19 \text{ kHz}$$

The Q value of 0.577 tells us that this is a Bessel response. With Eq. (21-21), the cutoff frequency is given by:

$$f_c = K_c f_p = 0.786(8.19 \text{ kHz}) = 6.44 \text{ kHz}$$

VCVS Equal Component low pass filter

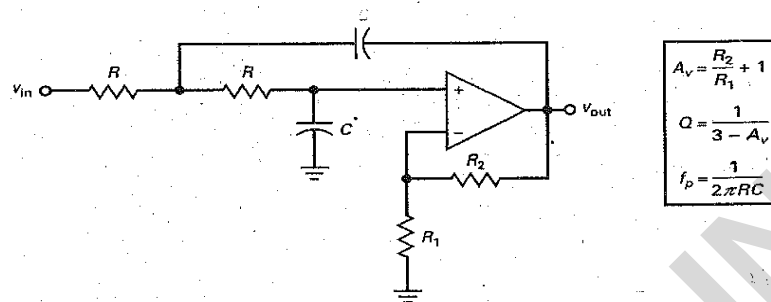
Figure 21-31 shows another Sallen-Key second-order low-pass filter. This time, both resistors and both capacitors have the same value. This is why the circuit is called a **Sallen-Key equal-component filter**. The circuit has a midband voltage gain of:

$$A_v = \frac{R_2}{R_1} + 1 \quad (21-29)$$

The operation of the circuit is similar to that of Sallen-Key unity-gain filter, except for the effect of the voltage gain. Since the voltage gain can produce more positive feedback through the feedback capacitor, the Q of the stage becomes a function of voltage gain and is given by:

$$Q = \frac{1}{3 - A_v} \quad (21-30)$$

Figure 21-31: VCVS equal-component stage.



After we calculate A_v , Q , and f_p with the equations shown in Fig. 21-31, the rest of the analysis is the same as before because a Butterworth filter has $Q = 0.707$ and $K_c = 1$. A Bessel filter has $Q = 0.577$ and $K_c = 0.786$. For

VCVS High pass filter

Figure 21-35a shows the Sallen-Key unity-gain high-pass filter and its equations. Notice that the positions of resistors and capacitors have been reversed. Also notice that Q depends on the ratio of resistances rather than capacitances. The calculations are similar to those discussed for low-pass filters, except that we have to divide the pole frequency by the K value. To calculate the cutoff frequency of a high-pass filter, we use:

$$f_c = \frac{f_p}{K_c} \quad (21-31)$$

Figure 21-35 Second-order VCVS high-pass stages: (a) Unity gain; (b) voltage gain greater than unity.

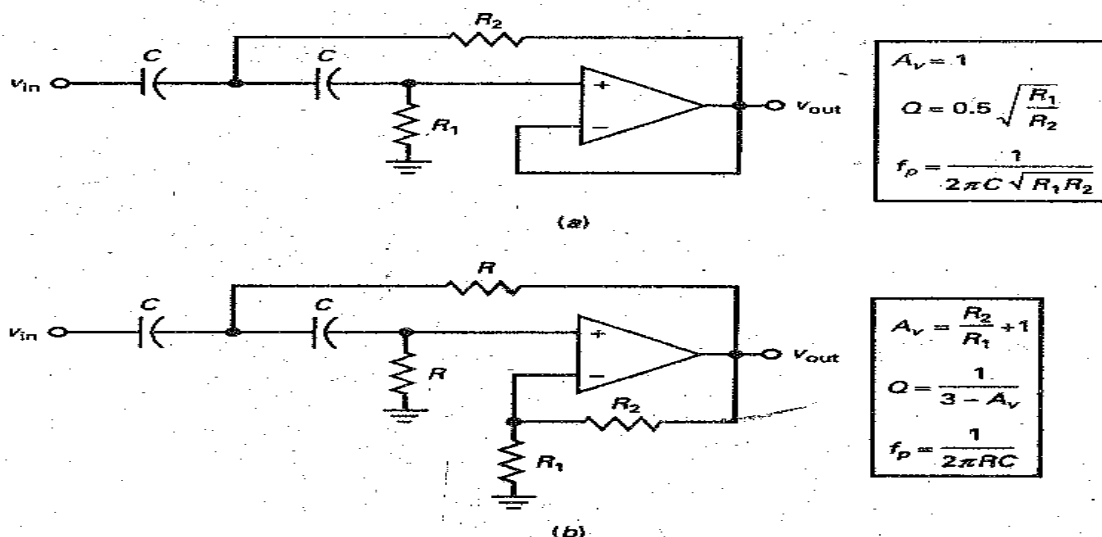


Figure 21-35b shows the Sallen-Key equal-component high-pass filter and its equations. All the equations are the same as for a low-pass filter. The positions of the resistors and capacitors have been reversed. The following examples show you how to analyze high-pass filters.

Example 21-9

III Multisim

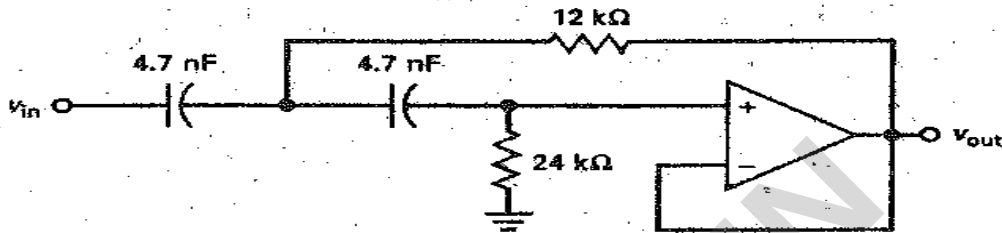
What are the pole frequency and Q of the filter shown in Fig. 21-36? What is the cutoff frequency?

SOLUTION The Q and pole frequency are:

$$Q = 0.5 \sqrt{\frac{R_1}{R_2}} = 0.5 \sqrt{\frac{24 \text{ k}\Omega}{12 \text{ k}\Omega}} = 0.707$$

$$f_p = \frac{1}{2\pi C \sqrt{R_1 R_2}} = \frac{1}{2\pi (4.7 \text{ nF}) \sqrt{(24 \text{ k}\Omega)(12 \text{ k}\Omega)}} = 2 \text{ kHz}$$

Figure 21-36 High-pass Butterworth example.



Since $Q = 0.707$, the filter has a Butterworth second-order response and:

$$f_c = f_p = 2 \text{ kHz}$$

The filter has a high-pass response with a break at 2 kHz, and it rolls off at 40 dB per decade below 2 kHz.

MFB(Multi feedback) Band pass filter

A bandpass filter has a center frequency and a bandwidth. Recall the basic equations for a bandpass response:

$$BW = f_2 - f_1$$

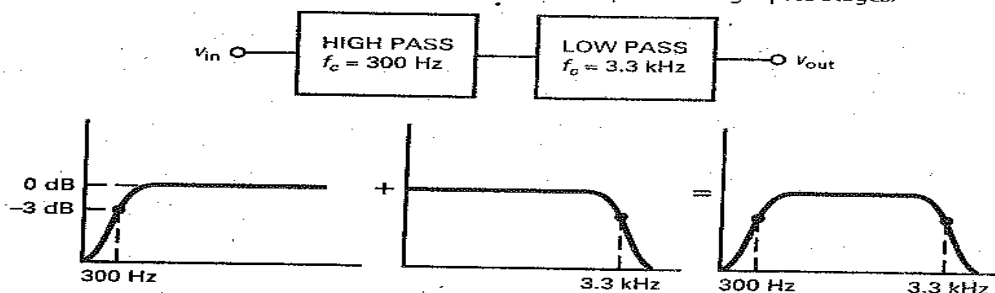
$$f_0 = \sqrt{f_1 f_2}$$

$$Q = \frac{f_0}{BW}$$

When Q is less than 1, the filter has a wideband response. In this case, a bandpass filter is usually built by cascading a low-pass stage with a high-pass stage. When Q is greater than 1, the filter has a narrowband response and a different approach is used.

Wide band filter

Figure 21-38 Wideband filter uses cascade of low-pass and high-pass stages.



Suppose we want to build a bandpass filter with a lower cutoff frequency of 300 Hz and an upper cutoff frequency of 3.3 kHz. The center frequency of the filter is:

$$f_0 = \sqrt{f_1 f_2} = \sqrt{(300 \text{ Hz})(3.3 \text{ kHz})} = 995 \text{ Hz}$$

The bandwidth is:

$$\text{BW} = f_2 - f_1 = 3.3 \text{ kHz} - 300 \text{ Hz} = 3 \text{ kHz}$$

Q is:

$$Q = \frac{f_0}{\text{BW}} = \frac{995 \text{ Hz}}{3 \text{ kHz}} = 0.332$$

Since Q is less than 1, we can use cascaded low-pass and high-pass stages, as shown in Fig. 21-38. The high-pass filter has a cutoff frequency of 300 Hz, and the low-pass filter has a cutoff frequency of 3.3 kHz. When the two decibel responses are added, we get a bandpass response with cutoff frequencies of 300 Hz and 3.3 kHz.

Narrow Band filter

When Q is greater than 1, we can use the **multiple-feedback (MFB)** filter shown in Fig. 21-39. First, notice that the input signal goes to the inverting input rather than the noninverting input. Second, notice that the circuit has two feedback paths, one through a capacitor and another through a resistor.

At low frequencies the capacitors appear to be open. Therefore, the input signal cannot reach the op amp, and the output is zero. At high frequencies the capacitors appear to be shorted. In this case, the voltage gain is zero because the feedback capacitor has zero impedance. Between the low and high extremes in frequency, there is a band of frequencies where the circuit acts like an inverting amplifier.

The voltage gain at the center frequency is given by:

$$A_v = \frac{-R_2}{2R_1} \quad (21-32)$$

This is almost identical to the voltage gain of an inverting amplifier: except for the factor of 2 in the denominator. The Q of the circuit is given by:

$$Q = 0.5 \sqrt{\frac{R_2}{R_1}} \quad (21-33)$$

which is equivalent to:

$$Q = 0.707 \sqrt{-A_v} \quad (21-34)$$

Figure 21-39 Multiple-feedback bandpass stage.

